



Trees

AIL7002

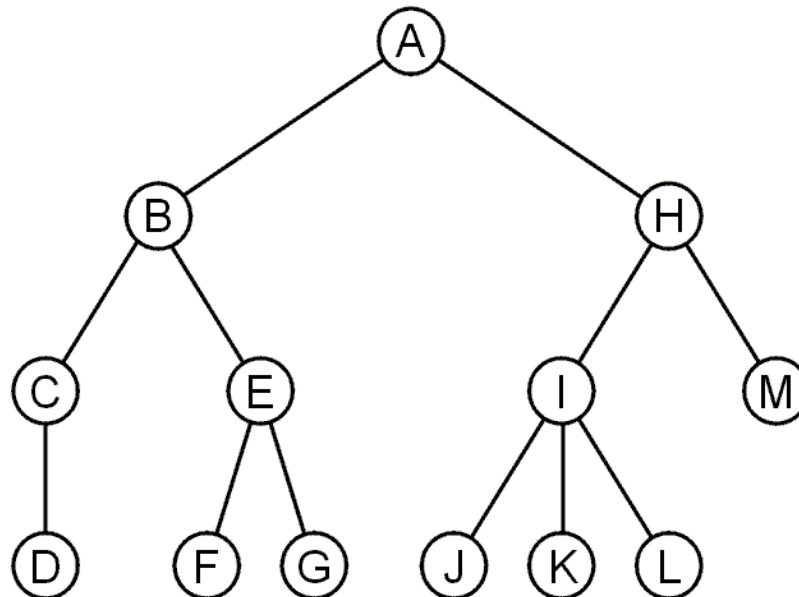
Acknowledgement :Many slides are courtesy
Douglas Harder, UWaterloo

Trees

A rooted tree data structure stores information in *nodes*

– Similar to linked lists:

- There is a first node, or *root*
- Each node has variable number of references to successors (children)
- Each node, other than the root, has **exactly one node as its predecessor** (or parent)

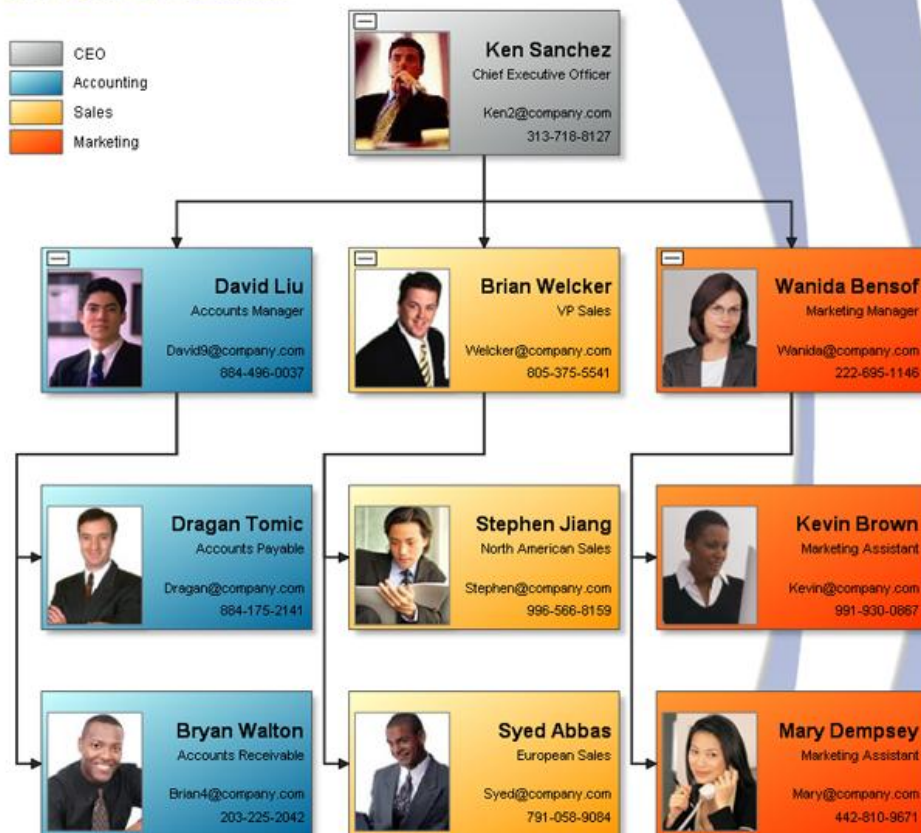


What are trees suitable for ?

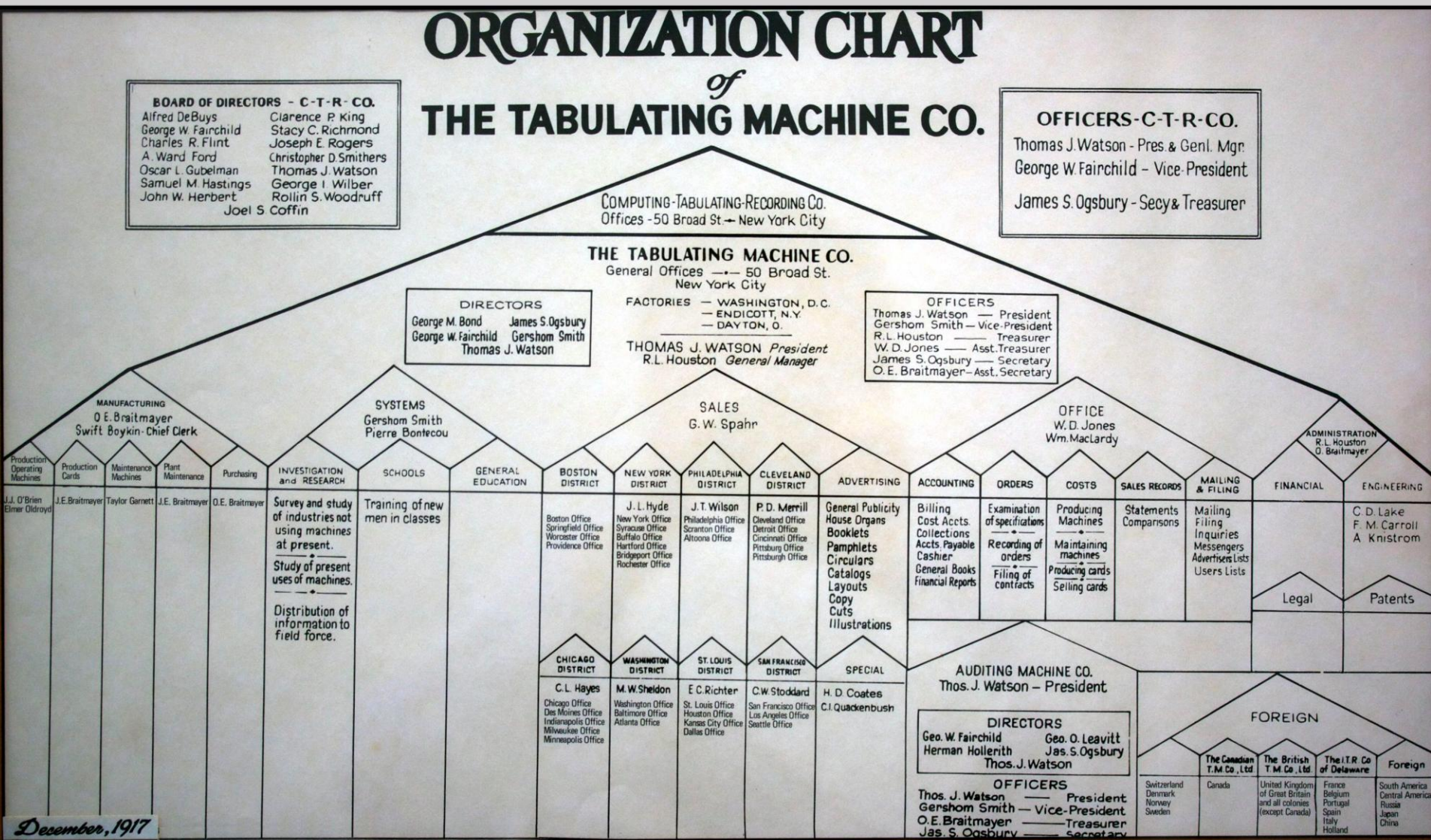
To store hierarchy of people

Company Org. Structure Contacts & Relations

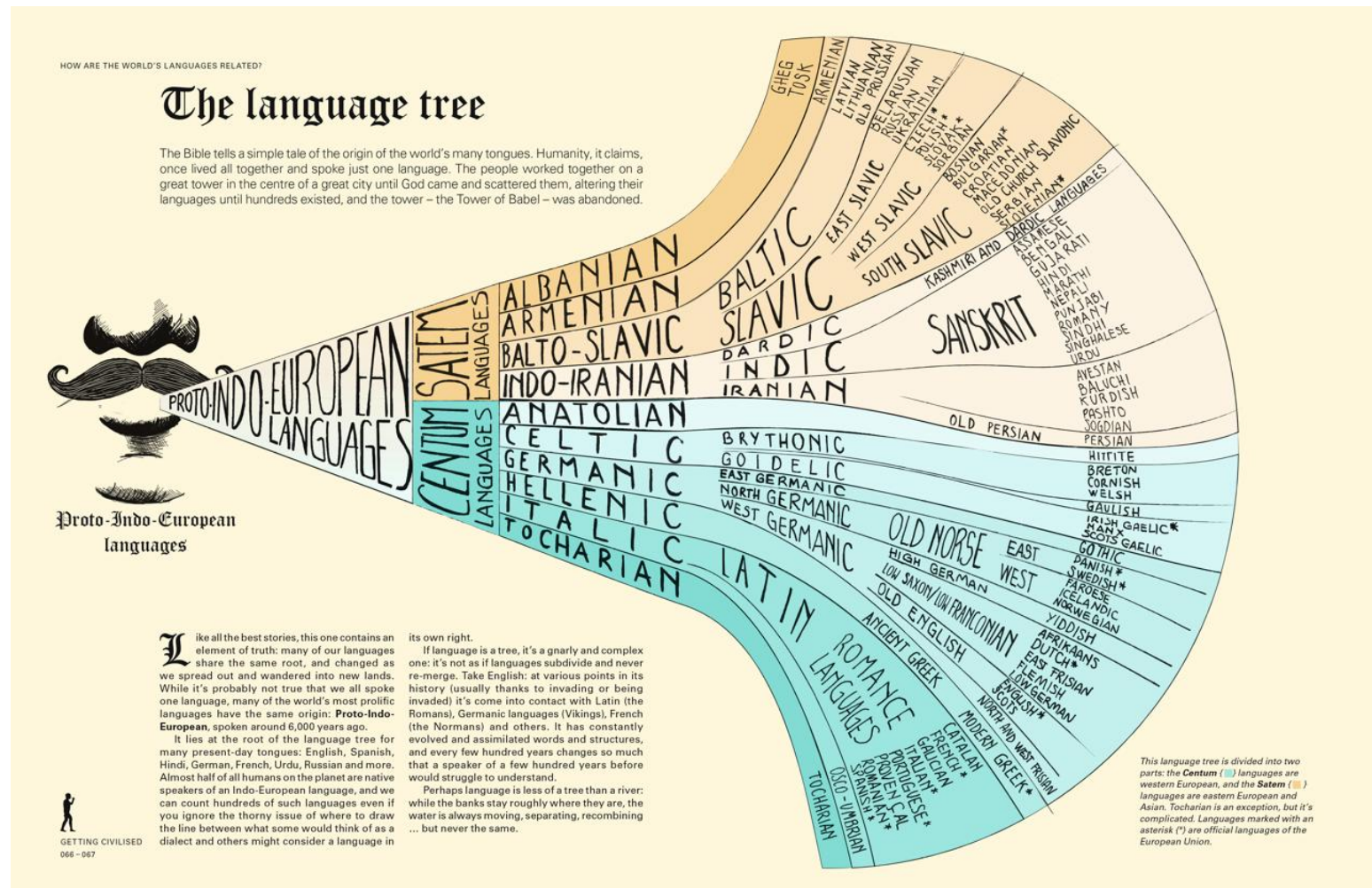
- CEO
- Accounting
- Sales
- Marketing



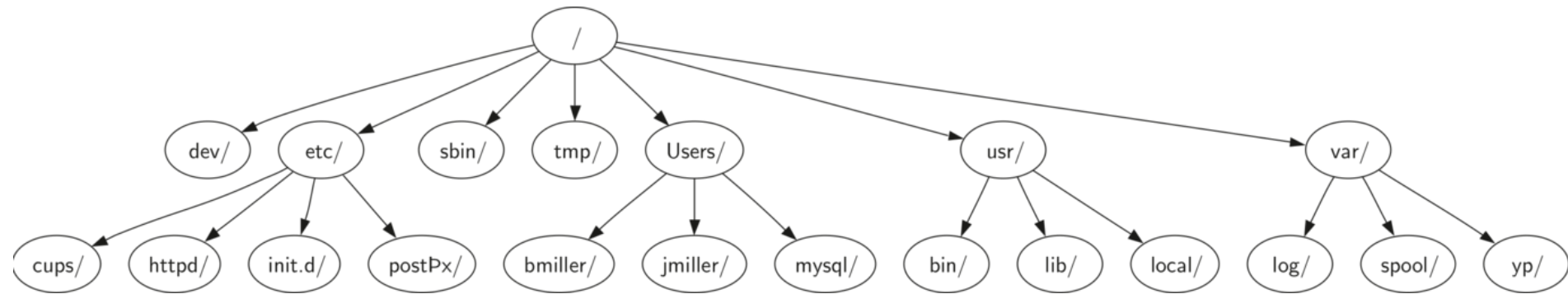
To store organization of departments



To capture the evolution of languages

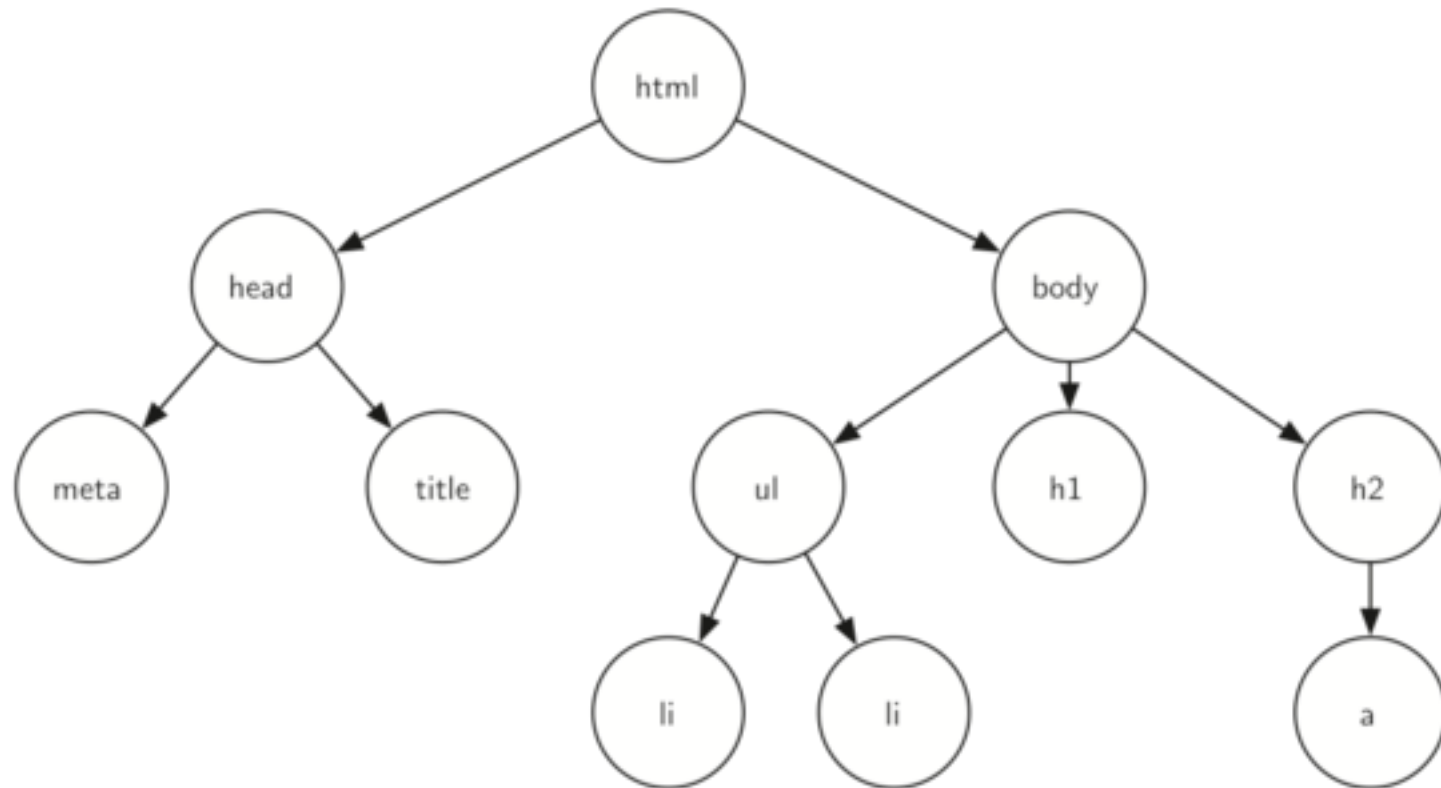


To organize file-systems



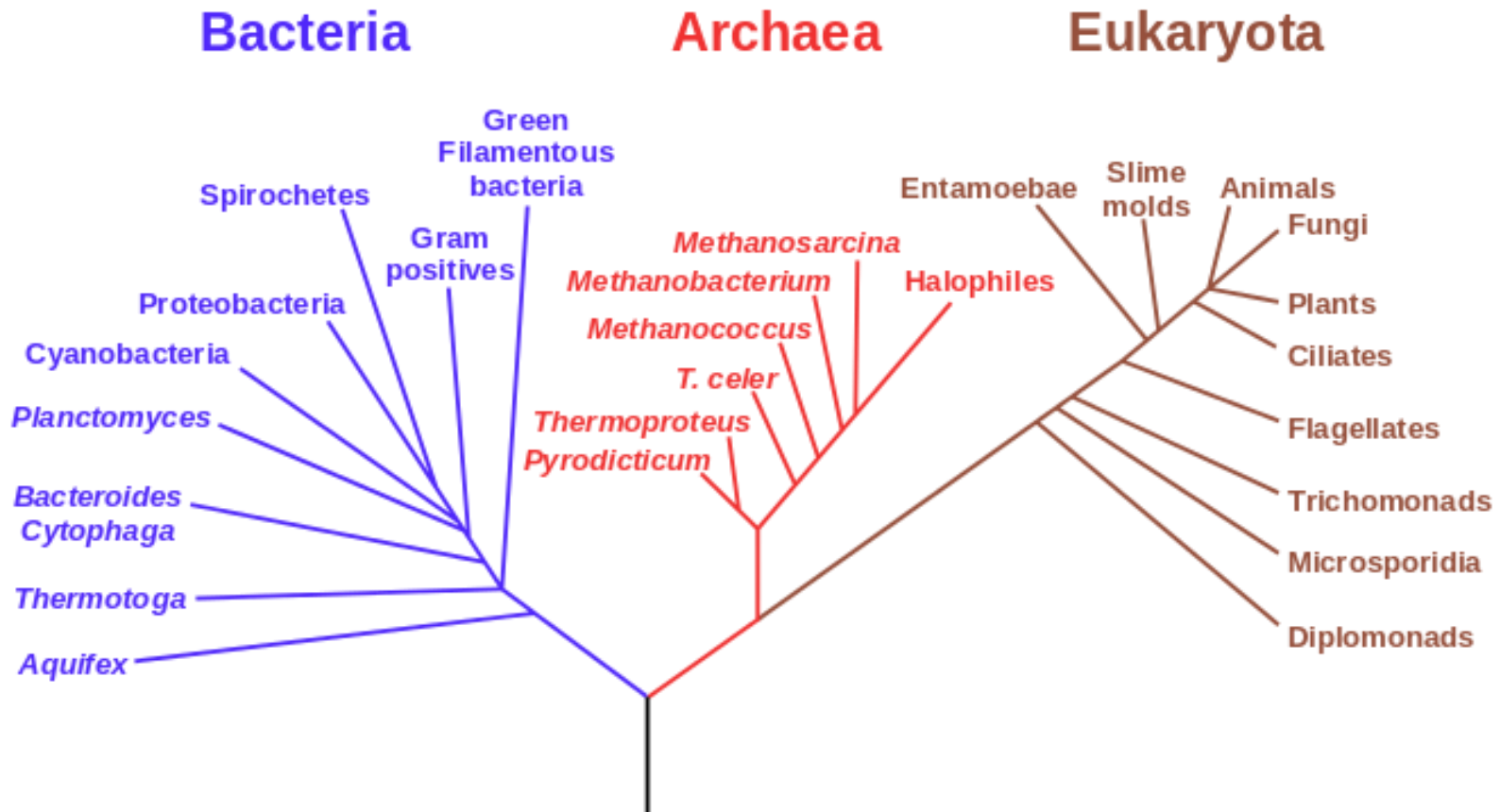
Unix file system

Markup elements in a webpage



To store phylogenetic data

Phylogenetic Tree of Life



This will be our running example. Will illustrate tree concepts using actual phylogenetic data.

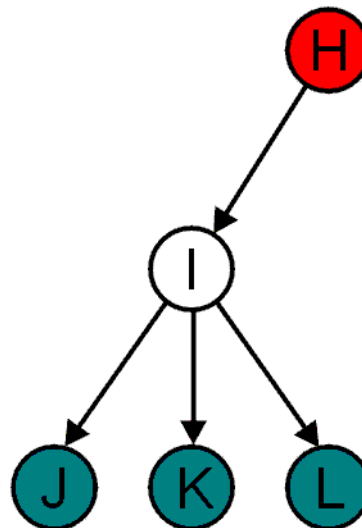
Terminology

All nodes will have zero or more child nodes or *children*

- I has three children: J, K and L

For all nodes other than the root node, there is one *parent* node

- H is the parent of I

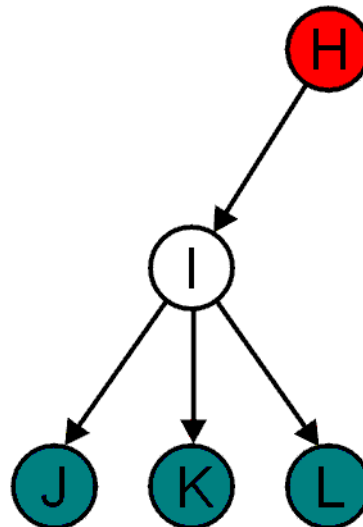


Terminology

The *degree* of a node is defined as the number of its children: $\deg(I) = 3$

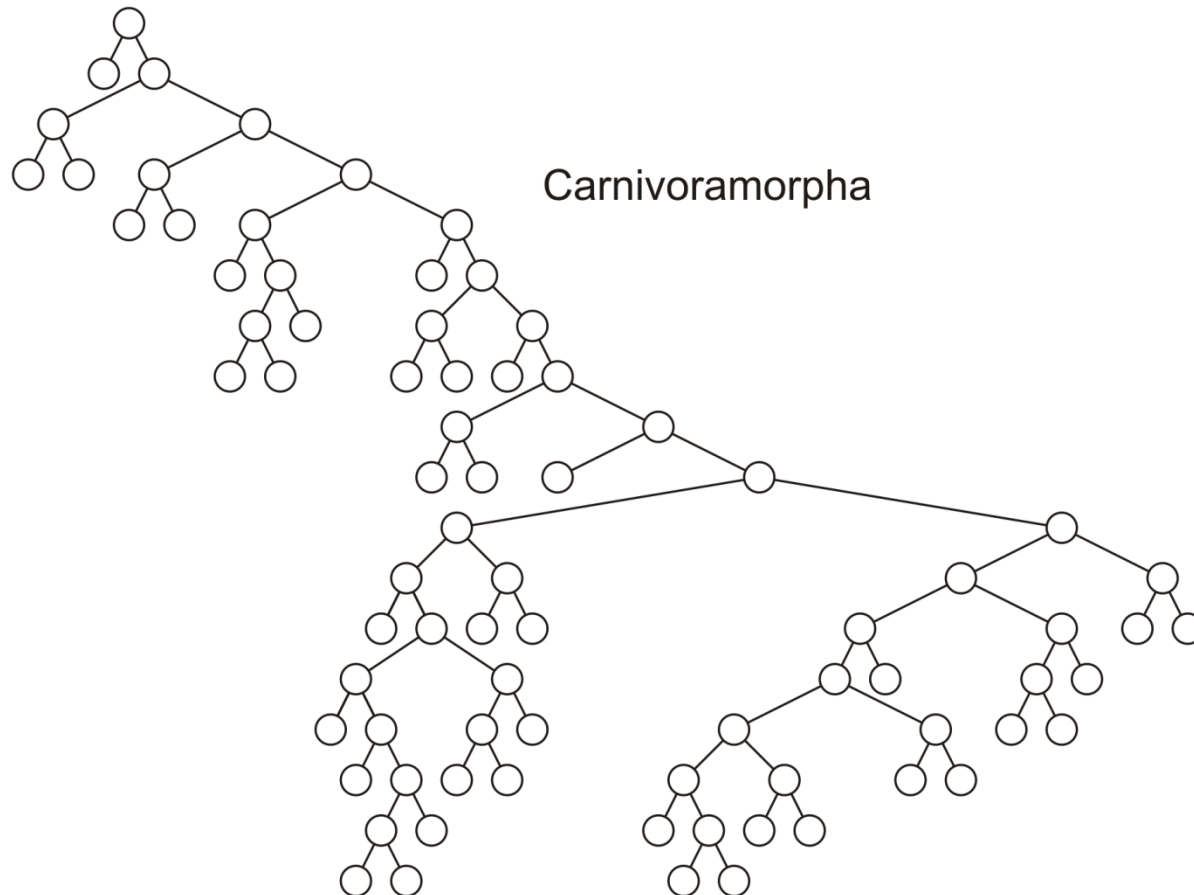
Nodes with the same parent are *siblings*

- J, K, and L are siblings



Terminology

Phylogenetic trees have nodes with **degree** 2 or 0:

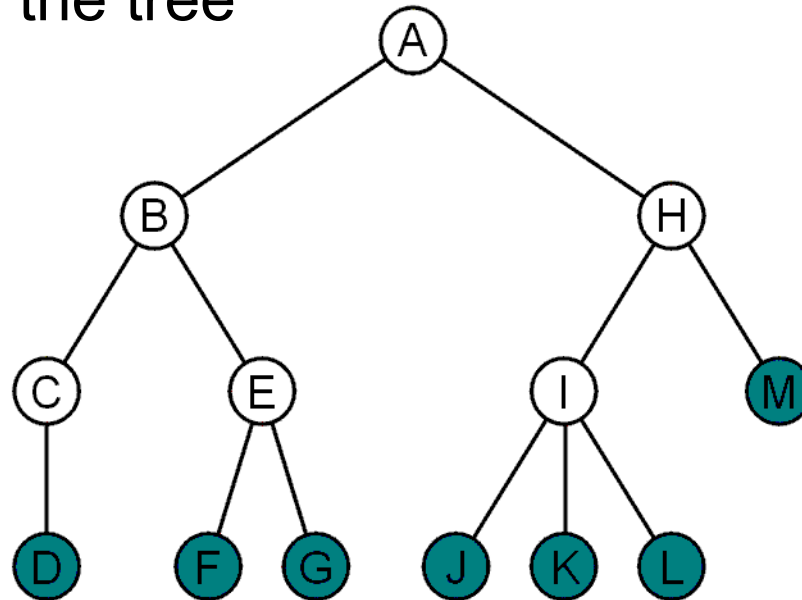


Wesley-Hunt, G. D.; Flynn, J. J. "Phylogeny of the Carnivora: basal relationships among the Carnivoramorphanes, and assessment of the position of 'Miacoidea'"

Terminology

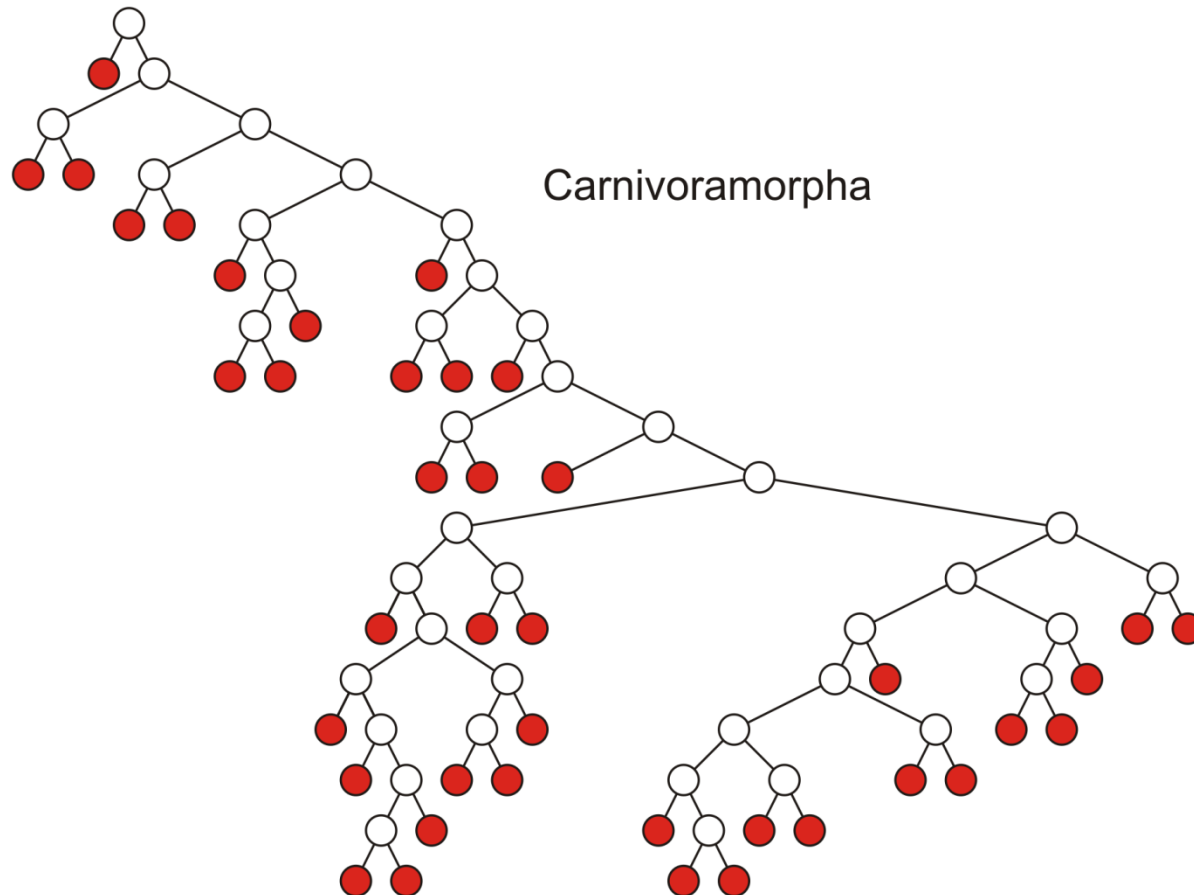
Nodes with degree zero are also called *leaf nodes*

All other nodes are said to be *internal nodes*, that is, they are internal to the tree



Terminology

Leaf nodes:



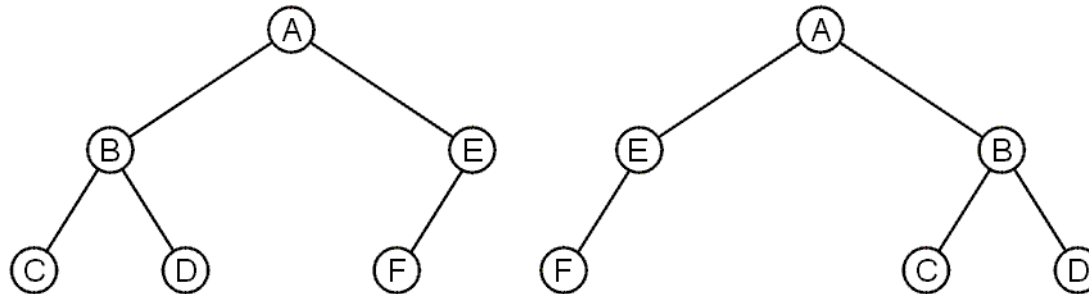
Wesley-Hunt, G. D.; Flynn, J. J. "Phylogeny of the Carnivora: basal relationships among the Carnivoramorpha, and assessment of the position of 'Miacoidea'"

Internal nodes:



Terminology

These trees are equal if the order of the children is ignored (*Unordered trees*)

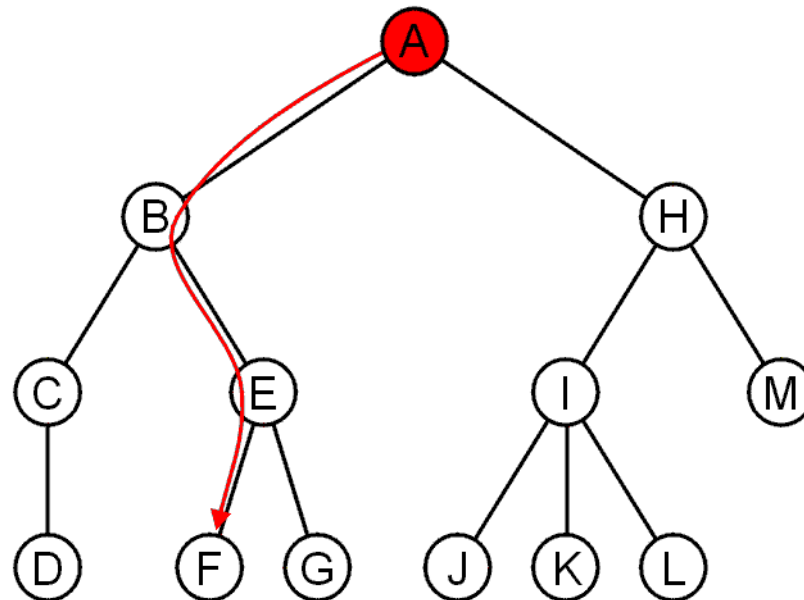


They are different if order is relevant (*ordered trees*)

- We will usually examine ordered trees (linear orders)

Terminology

The shape of a rooted tree gives a natural flow from the *root node*, or just *root*



Terminology

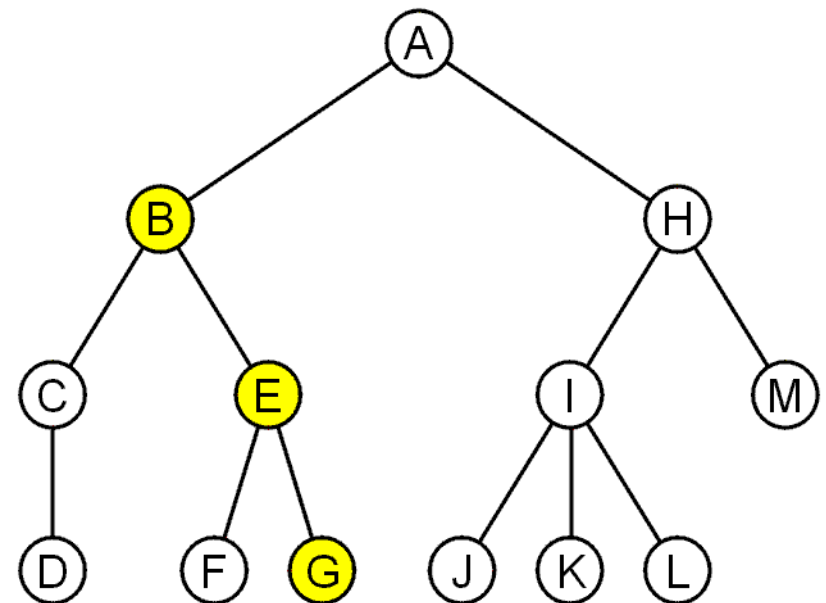
A **path** is a sequence of nodes

$$(a_0, a_1, \dots, a_n)$$

where a_{k+1} is a child of a_k is

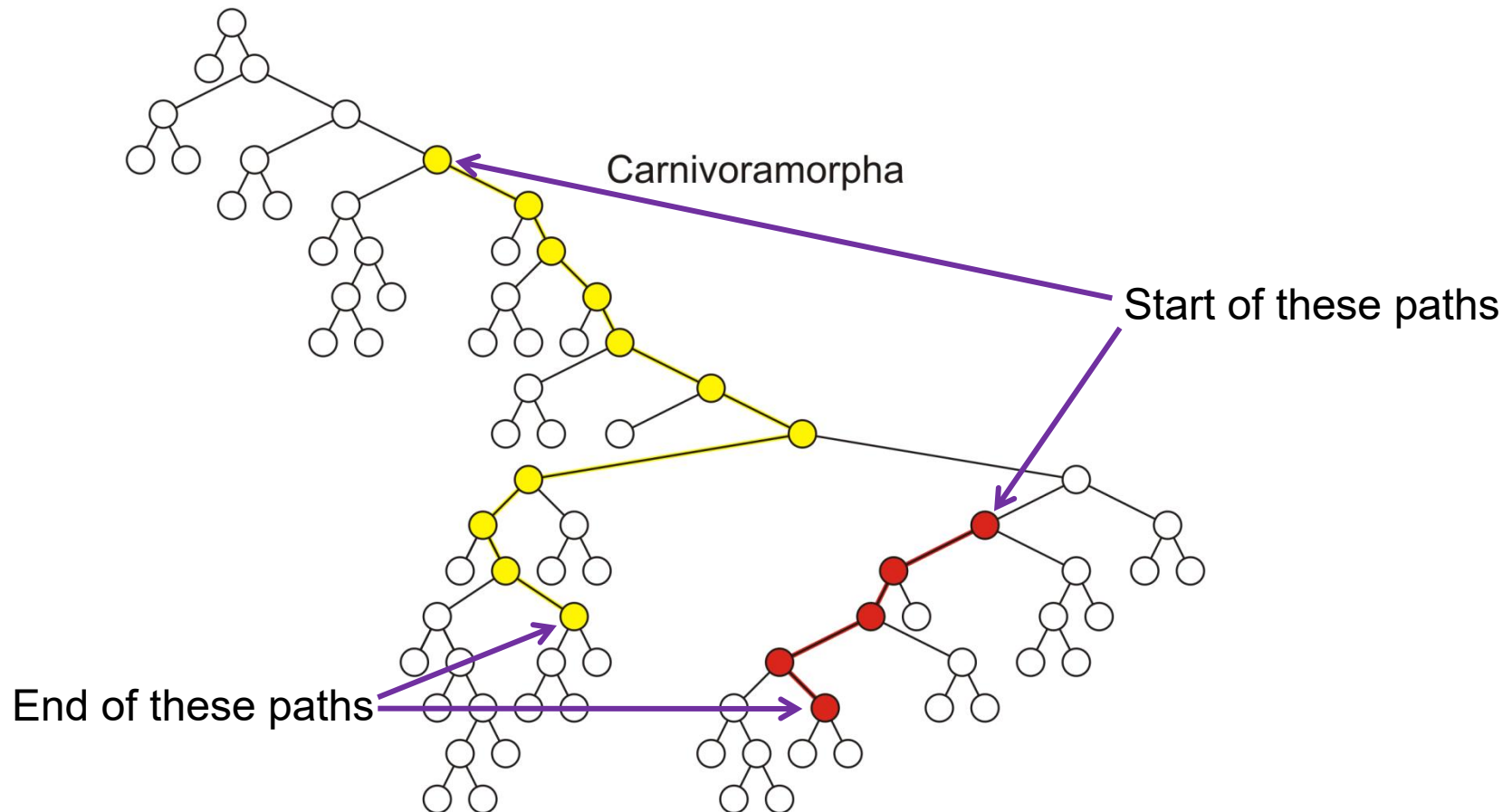
The **length** of this path is n

E.g., the path (B, E, G)
has length 2



Terminology

Paths of length 10 (11 nodes) and 4 (5 nodes)



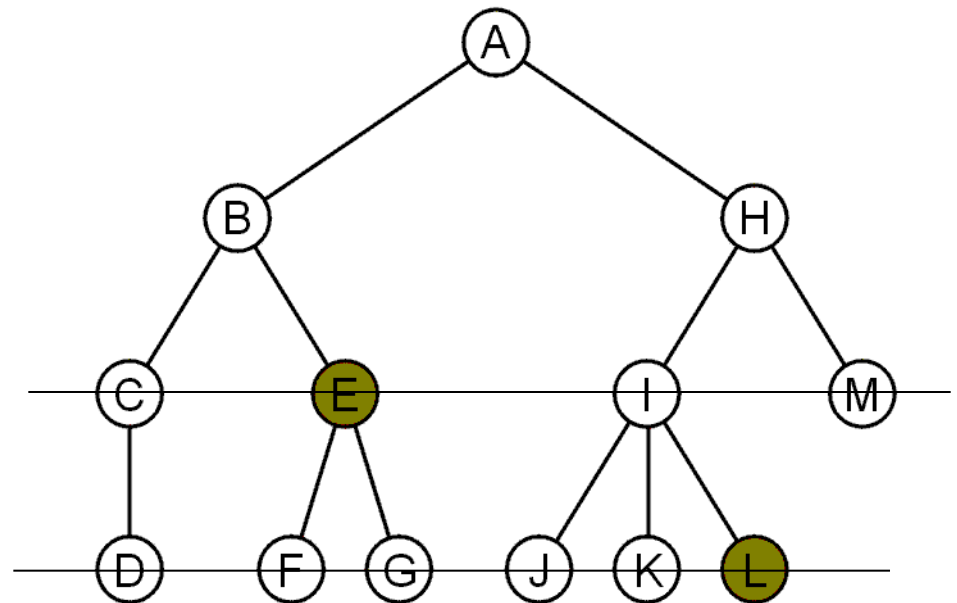
Wesley-Hunt, G. D.; Flynn, J. J. "Phylogeny of the Carnivora: basal relationships among the Carnivoramorphans, and assessment of the position of 'Miacoidea'"

Terminology

For each node in a tree, there exists a unique path from the root node to that node

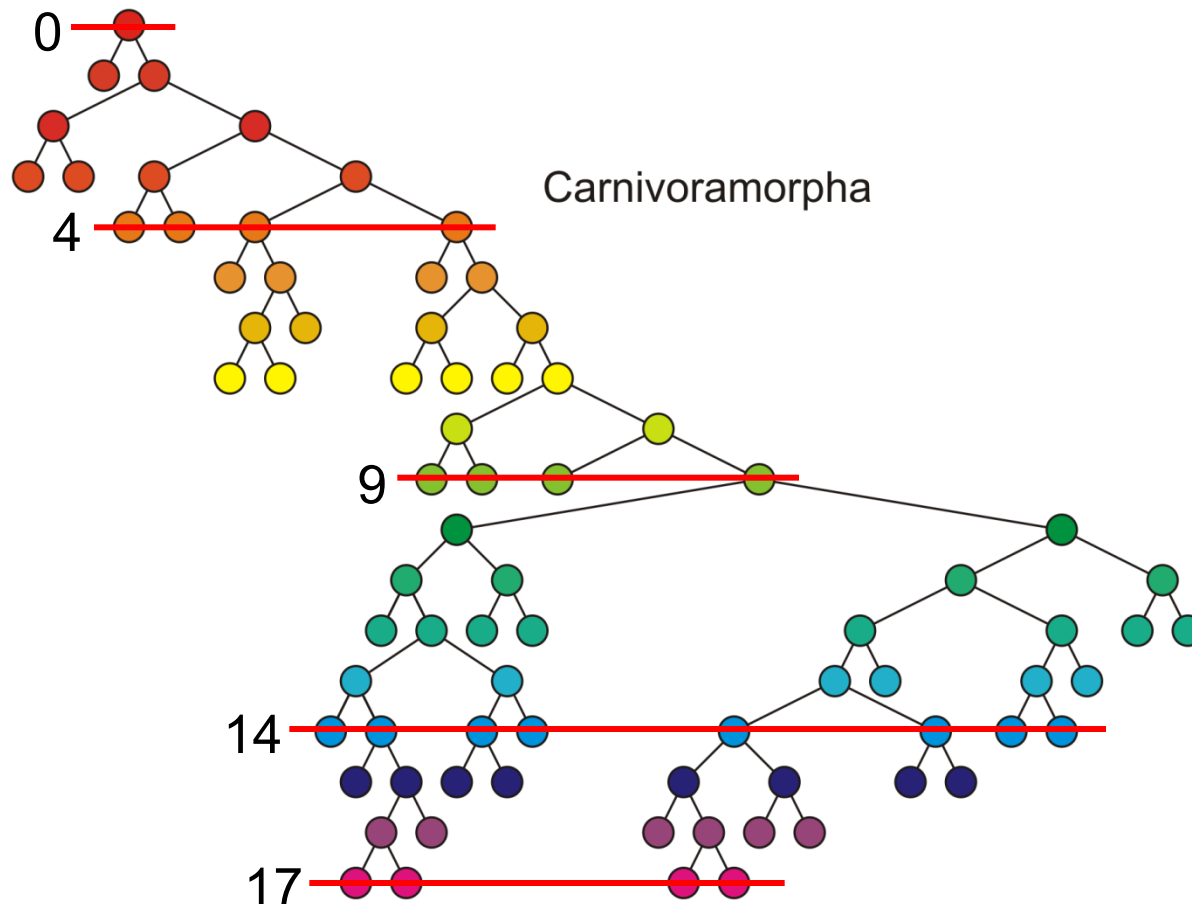
The length of this path is the *depth* of the node, e.g.,

- E has depth 2
- L has depth 3



Terminology

Nodes of depth up to 17



Wesley-Hunt, G. D.; Flynn, J. J. "Phylogeny of the Carnivora: basal relationships among the Carnivoramorphans, and assessment of the position of 'Miacoidea'"

Terminology

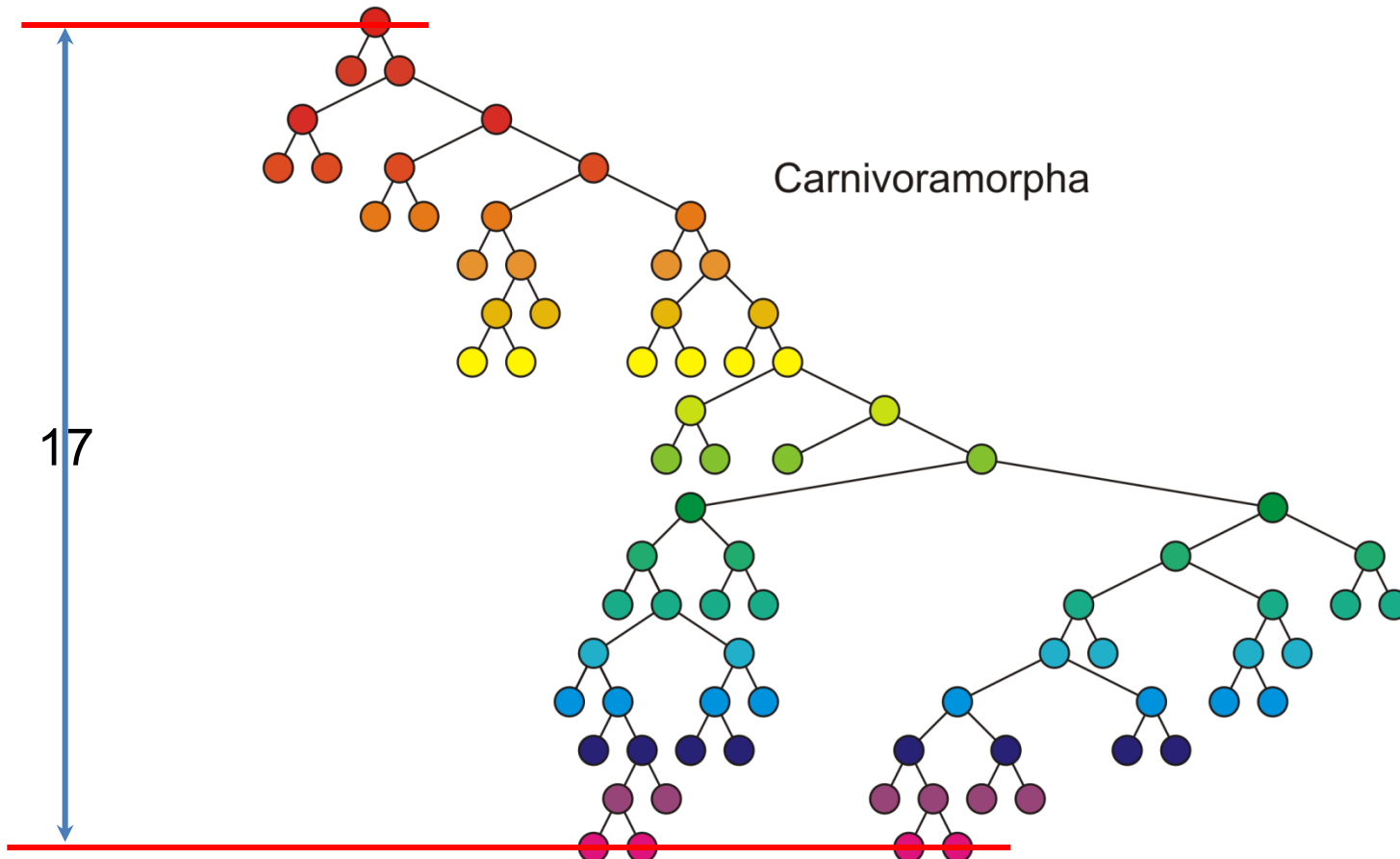
The *height* of a tree is defined as the maximum depth of any node within the tree

The height of a tree with one node is 0
– Just the root node

For convenience, we define the height of the empty tree to be -1

Terminology

The height of this tree is 17



Wesley-Hunt, G. D.; Flynn, J. J. "Phylogeny of the Carnivora: basal relationships among the Carnivoramorphans, and assessment of the position of 'Miacoidea'"

Terminology

If a path exists from node a to node b :

- a is an *ancestor* of b
- b is a *descendant* of a

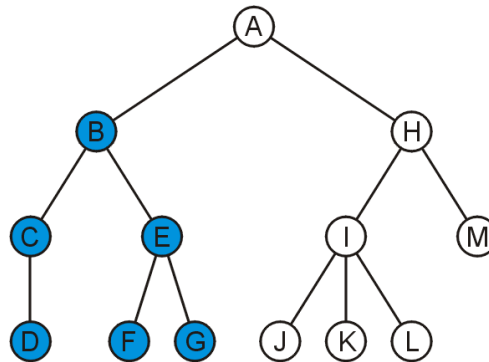
Thus, a node is both an ancestor and a descendant of itself

- We can add the adjective *strict* to exclude equality: a is a *strict descendant* of b if a is a descendant of b but $a \neq b$

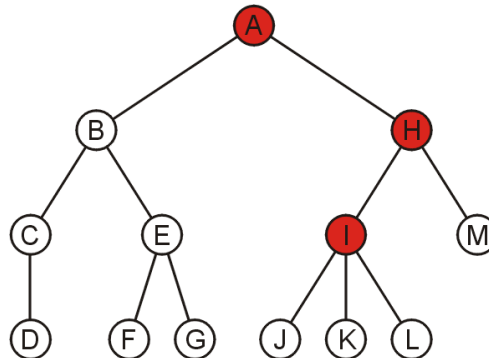
The root node is an ancestor of all nodes

Terminology

The **descendants** of node B are B, C, D, E, F, and G:

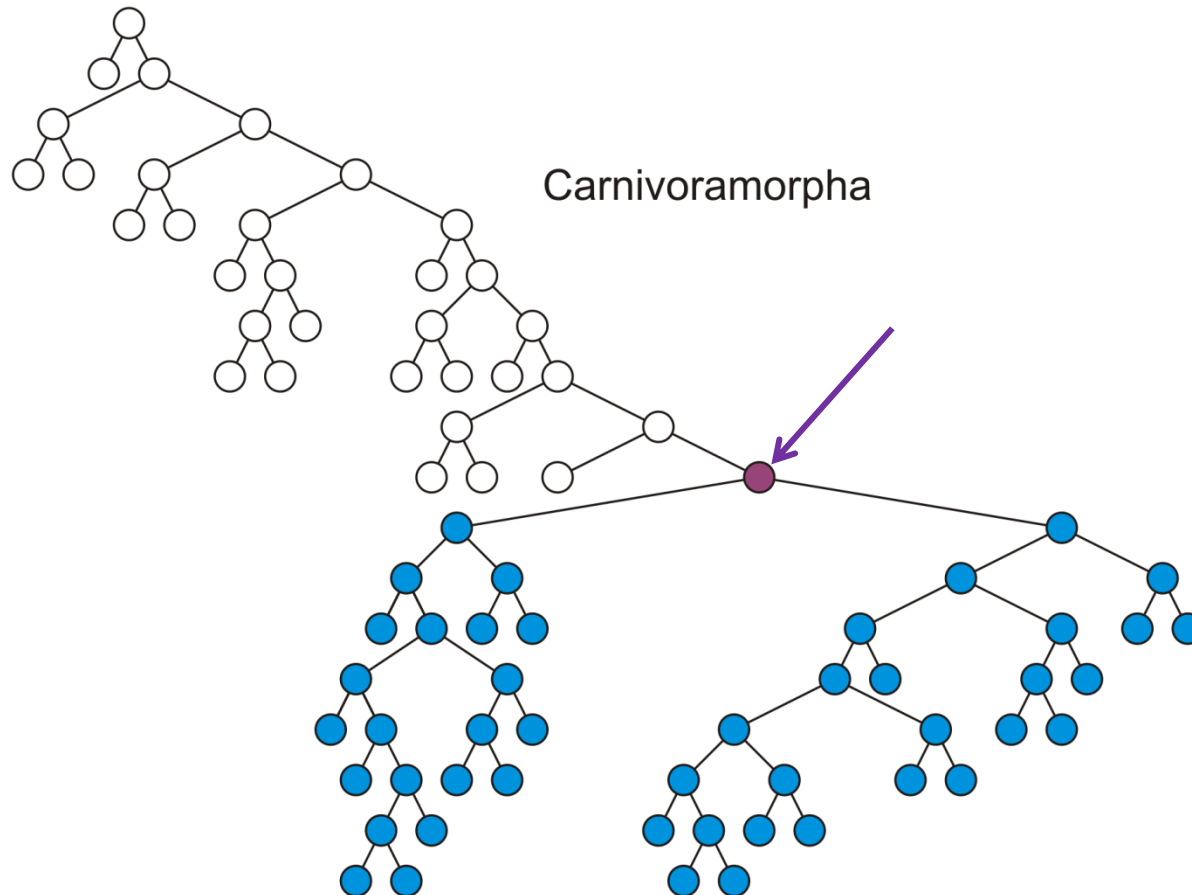


The **ancestors** of node I are I, H, and A:



Terminology

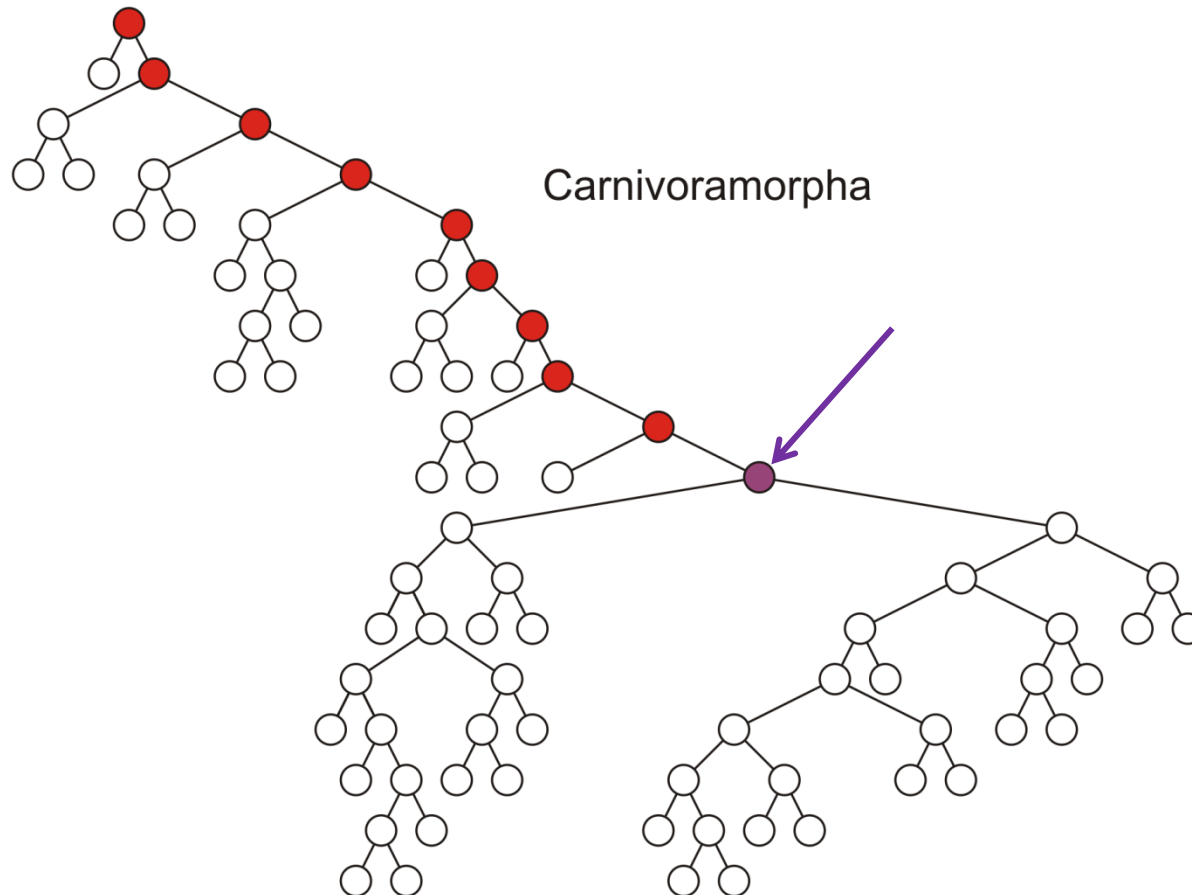
All **descendants** (including itself) of the indicated node



Wesley-Hunt, G. D.; Flynn, J. J. "Phylogeny of the Carnivora: basal relationships among the Carnivoramorphans, and assessment of the position of 'Miacoidea'"

Terminology

All **ancestors** (including itself) of the indicated node



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